MILITARY SPECIFICATION

SPACER, GRAIN
M753 8 INCH PROJECTILE

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for grain spacers used in M753, 8 Inch Projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-R-6130	Rubber, Cellular, Chemically Blown
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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STANDARDS

MILITARY

MIL-STD-286	Propellant, Solid, Sampling, Examination and Tests
MIL-STD-1188	Commercial Packaging of Supplies and Equipment

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.6)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9328351	Spacer, Grain
9328353	Sponge Rubber, Silicone (Medium Grade)
11828869	Propellant Grain

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 692	Specific Gravity and Density of Plastics by Displacement
ASTM D 2240	Rubber Property, Durometer Hardness Test for

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Construction. The grain spacer shall comply with all requirements specified on drawing (dwg.) 9328351, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the dwg. and herein.

3.2 Materials. Silicone sponge rubber used in fabricating grain spacers shall comply with all requirements specified on dwg. 9328353 and with all requirements specified herein. The contractor shall submit test samples of the materials to the procuring agency for analysis and approval in accordance with the requirements of this specification prior to use in fabrication of grain spacers (see 6.3).

3.3 Material properties.

3.3.1 Physical properties. The physical properties of the silicone sponge rubber at 73 ± 3°F (23 ± 3°C) shall be as specified in Table I. Testing shall be performed in accordance with 4.5.1 thru 4.5.3, as applicable.

3.3.2 Chemical composition. The silicone rubber shall contain a minimum of 55% organic material by weight. Testing shall be performed in accordance with 4.5.4.

3.3.3 Infrared spectrum. The infrared spectrum of the silicone sponge rubber shall qualitatively match the wavelength characteristics of Figure 2. Testing shall be performed in accordance with 4.5.5.

TABLE I. Physical properties

Properties	
Density at 73°F, max. (lb/cu. in)	.020
Hardness, Shore A	6-8
Compression deflection, psi (compressed at 75% of original thickness at room temp.)	6-14
Compression set, % max (compressed to 50% of original thickness for 22 hours at 212°F)	15

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3.3.4 Transition temperature. The transition temperature of the silicone rubber (the temperature at which the material changes from a rubber to a rigid solid) shall not be above -60°F (-52°C). Testing shall be performed in accordance with 4.5.6.

3.3.5 Reactivity. The reactivity between the silicone sponge rubber and AHH propellant shall not exceed 1.0 milliliter (ml) when tested in accordance with 4.5.7.

3.4 First article. This specification contains technical provisions for the first article inspection (see 4.3). Requirements for submission of first article samples shall be as specified in the contract.

3.5 Workmanship. Grain spacers shall be fabricated in a thoroughly workmanlike manner and shall be free of large voids or any other obvious material defects. Finished parts shall be clean and free of dirt, grease, contamination or any other foreign matter. Cleaning agents shall not be injurious to the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract or purchase order, the provisions of MIL-A-48078 shall apply and and are hereby made a part of this detail specification.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. When the contract requires first article inspection (see 3.4), the contractor shall

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submit a first article sample as designated by the contracting officer for evaluation in accordance with 4.3.2. The sample shall consist of the following items in the sample quantities as indicated.

Item	Drawing and Part No.	Quantity
Specimen of cured silicone sponge rubber for material property analysis	9328353	1 square foot
Grain spacers	9328351	10

4.3.2 Inspections to be performed. The first article samples submitted in accordance with 4.3.1 shall be subjected to any or all of the requirements of the applicable dwgs. and to the inspections and test specified in the first article inspection table herein. All parts shall have been manufactured using the same production process as will be used in fulfilling the contract and all materials shall be obtained from the same source as will be used in regular production and approved in accordance with 3.2. If the first article sample fails to meet any of the applicable requirements, the first article sample shall be rejected.

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FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63172(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.1	Spacer, Grain	1 OF 1	9328351 and 9328353
			NEXT HIGHER ASSEMBLY: 11828869

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Classificiation of Defects and Tests Silicone Sponge Rubber Properties	(a)			4.4.2.1
	Classification of Defects and Tests Grain Spacers	10			4.4.2.2

NOTES:

(a) One square foot specimen of cured silicone sponge rubber.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.3.3 Inspection equipment. The inspections and tests specified in 4.3.2 shall be accomplished with the gages and test equipment specified in 4.4.4.

4.3.4 Rejection. See MIL-A-48078

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots of finished cushions shall comply with the lot formation provisions of MIL-A-48078. In addition, an inspection lot shall consist of items fabricated from a single inspection lot of material produced by a single manufacturer in one continuous process employing not more than one lot of base material and curing agent approved in accordance with 3.2.

4.4.2 Examinations and tests. See MIL-A-48078 and the quality conformance inspection tables herein. If any sample or specimen fails to meet any of the applicable requirements, the inspection lot which it represents shall be rejected.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63172(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Silicone Sponge Rubber Properties				9328353
					NEXT HIGHER ASSEMBLY
					11828869
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Physical properties	(a)		3.3.1	4.5.1 thru 4.5.3
102.	Chemical composition	(a)		3.3.2	4.5.4
103.	Infrared spectrum	(a)		3.3.3	4.5.5
104.	Transition temperature	(a)		3.3.4	4.5.6
105.	Reactivity	(a)		3.3.5	4.5.7

NOTES:

(a) One square foot specimen of cured silicone rubber will be required to perform material property tests (see 6.3).

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63172(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Spacer, Grain	1 OF 1	9328351
			NEXT HIGHER ASSEMBLY
			11828869

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Length	4.4.3.1		3.1	Gage/TMC
102.	Width	4.4.3.1		3.1	Gage/TMC
103.	Thickness	4.4.3.1		3.1	Gage
Minor:					
201.	Poor workmanship		100%	3.5	Visual

NOTES:

(a) Gage each sheet of material when sheet material is used or one out of every ten grain spacers if other fabrication methods are used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63172(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Packaging	1 OF 1	See 5.0
			NEXT HIGHER ASSEMBLY: N.A.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Packaging and packing not in accordance with applicable requirements	1/lot		5.1	Visual
102.	Containers or packages not securely sealed	1/lot		5.1	Visual/Manual
103.	Marking missing, incorrect or illegible	1/lot		5.2	Visual
104.	Poor workmanship	1/lot		---	Visual

NOTES:

10

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing. Testing is described in the first article and quality conformance inspection tables and herein.

4.4.3.1 Tool manufacturing control. When tool manufacturing control (TMC) is specified, the first five pieces initially produced from each mold or die inspected for the applicable classification of defects herein shall constitute initial tool acceptance. Subsequently, the first and last five pieces from each days production inspected for the applicable classification of defects herein shall constitute inspection lot acceptance. In addition, any time the tool requires repair an additional five pieces from each tool shall be inspected for the applicable classification of defects herein before resumption of production. If any sample fails to meet any of the applicable inspection requirements, the lot of production under consideration shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" columns in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see section 6 of MIL-A-48078 and 6.4 herein).

4.5 Test methods and procedures.

4.5.1 Density (see 3.3.1). The density of the silicone sponge rubber shall be determined in accordance with ASTM D-792, Method I or an approved equivalent.

4.5.2 Hardness (see 3.3.2). The Shore A Durometer hardness of the silicone sponge rubber shall be measured in accordance with ASTM D-2240 or an approved equivalent.

4.5.3 Compression deflection and set (see 3.3.1). The compression deflection and set (recovery) of the silicone sponge rubber shall be determined in accordance with 4.6.1.2 and 4.6.1.3 of MIL-R-6130 or approved equivalents.

4.5.4 Chemical composition (see 3.3.2). The organic content of the silicone sponge rubber shall be determined at 650°C (the temperature at which most of the decomposable matter is lost) using a Dupont 950 Thermogravimetric Analyzer or an approved equivalent to measure and record the sample weight loss as a function of temperature as shown in Figure 1. The temperature change shall be 10°C/min. and the medium shall be air.

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4.5.5 Infrared spectrum (see 3.3.3). The infrared spectrum of the silicone sponge rubber shall be determined by pyrolyzing a sample in a test tube over a bunsen burner, transferring the resultant liquid condensate into an infrared transmitting window and recording the infrared spectrum with an infrared spectrophotometer as shown in Figure 2.

4.5.6 Transition temperature (see 3.3.4). The transition temperature of the silicone sponge rubber shall be determined using a Dupont 941 Thermomechanical Analyzer or an approved equivalent to continuously measure and record the expansion of the material as a function of temperature as shown in Figure 3. The temperature change shall be 5°C/min. and the medium shall be air.

4.5.7 Reactivity (see 3.3.5). Reactivity tests shall be performed in accordance with MIL-STD-286, Method 408.1 or by Modified Taliani Test with apparatus described in MIL-STD-286, Method 406.1.2.

5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified in the contract or order, grain spacers shall be packaged, packed and marked in accordance with MIL-STD-1188 (commercial pack).

6. NOTES

6.1 Intended use. The grain spacers covered by this specification are intended for use in fabrication of propellant grains for rocket motors of XM753 8 Inch Projectiles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents would specify the following:

a. Title, number and date of this specification.
b. First article inspection, when required (see 3.4).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the

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contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirements	Applicable DID no.
4.4.4	Inspection equipment approval	DI-R-1714 (Tailored)

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Test samples. Material samples shall be submitted for approval to Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-LCA-OP, Dover, New Jersey 07801 or an approved laboratory.

6.4 Equipment design approval. Submit equipment designs, as required, to Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.5 Supersession data. This specification includes the requirements of AR-PD-21 dated 1 November 1978 with Amendmend 1 dated 10 November 1980.

6.6 Drawings and specifications. Drawings and specifications listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Command may also include dwgs. and specifications prepared and identified as Frankford Arsenal, Edgewood Arsenal, Rock Island Arsenal or Picatinny Arsenal dwgs. and specifications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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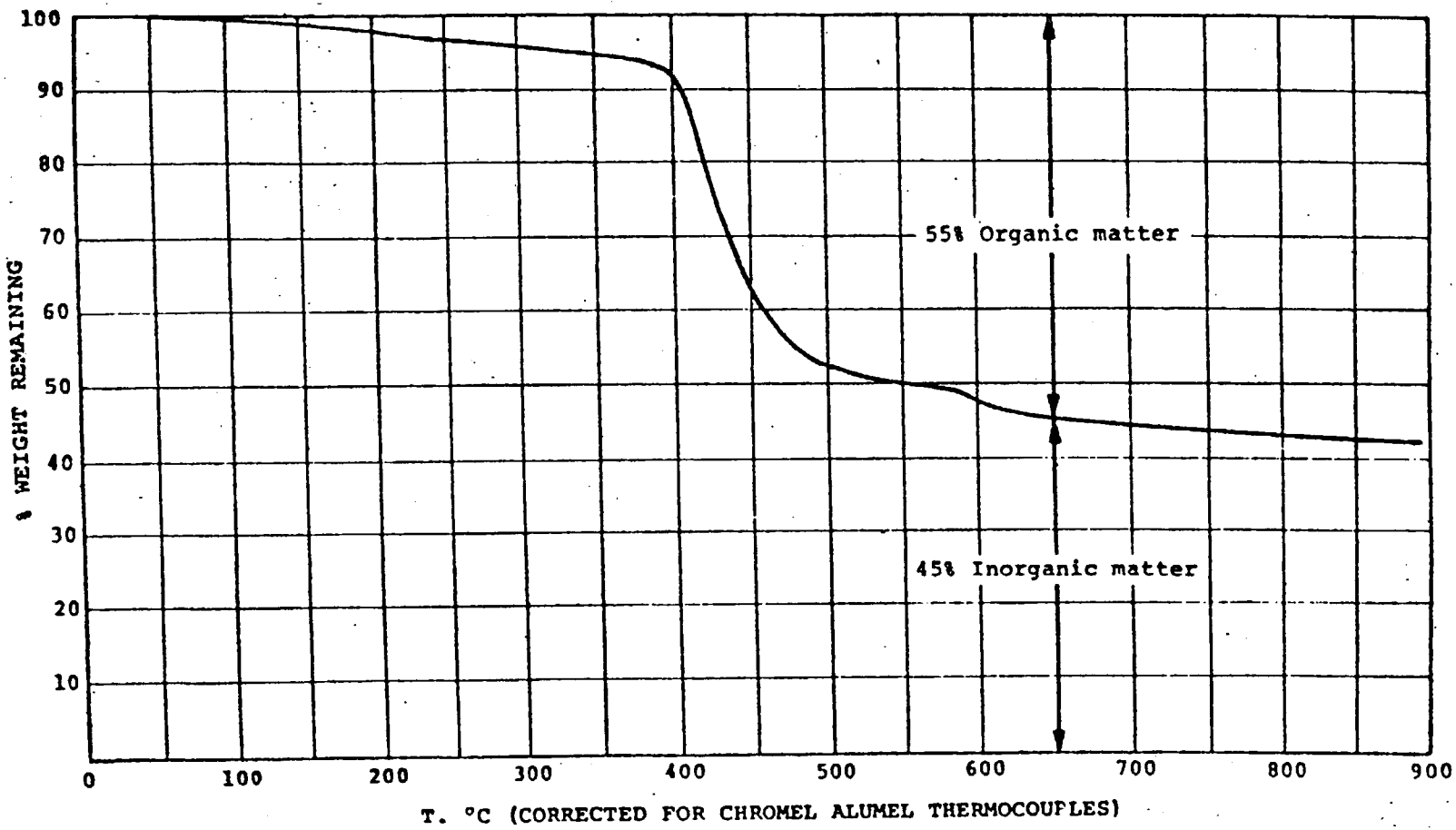
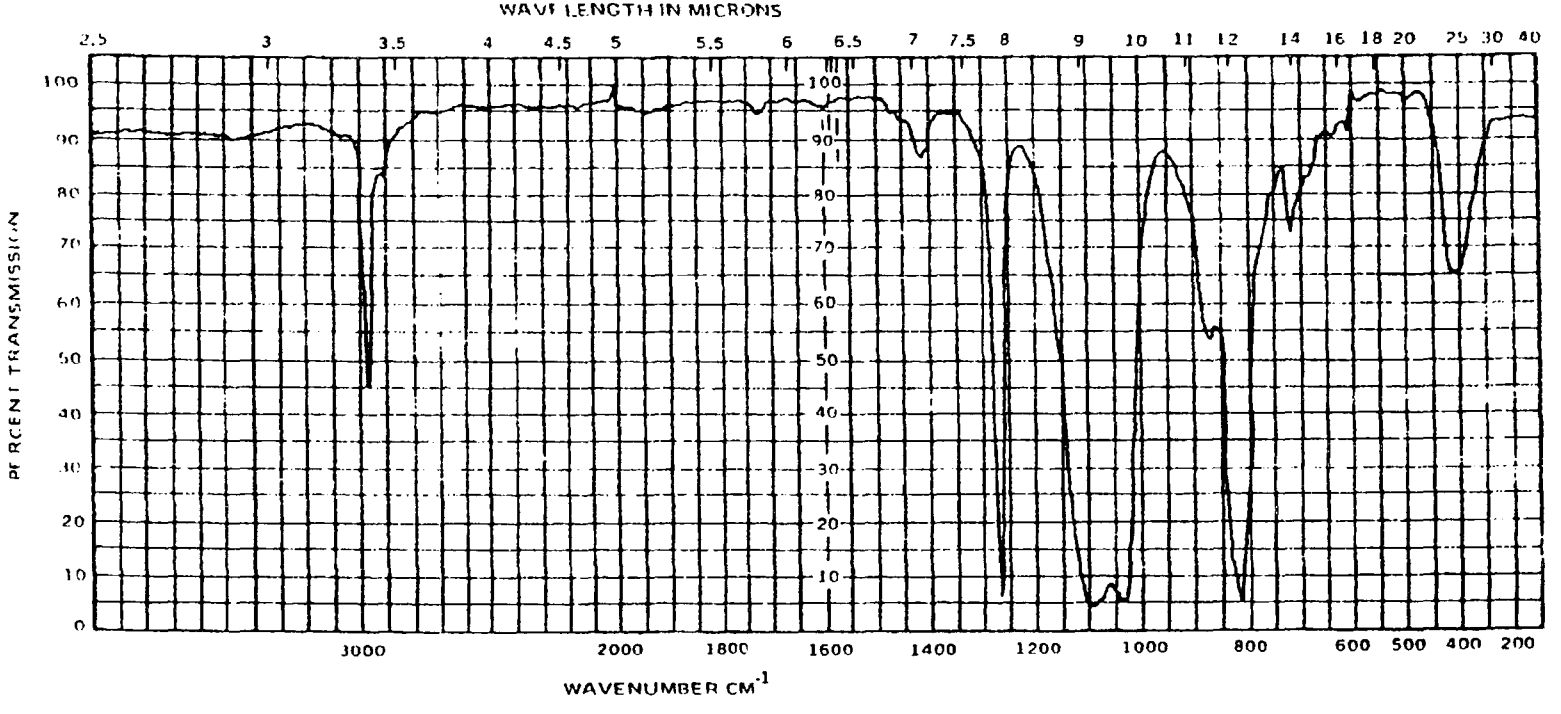


FIGURE 1. Chemical Composition (Medium Grade).

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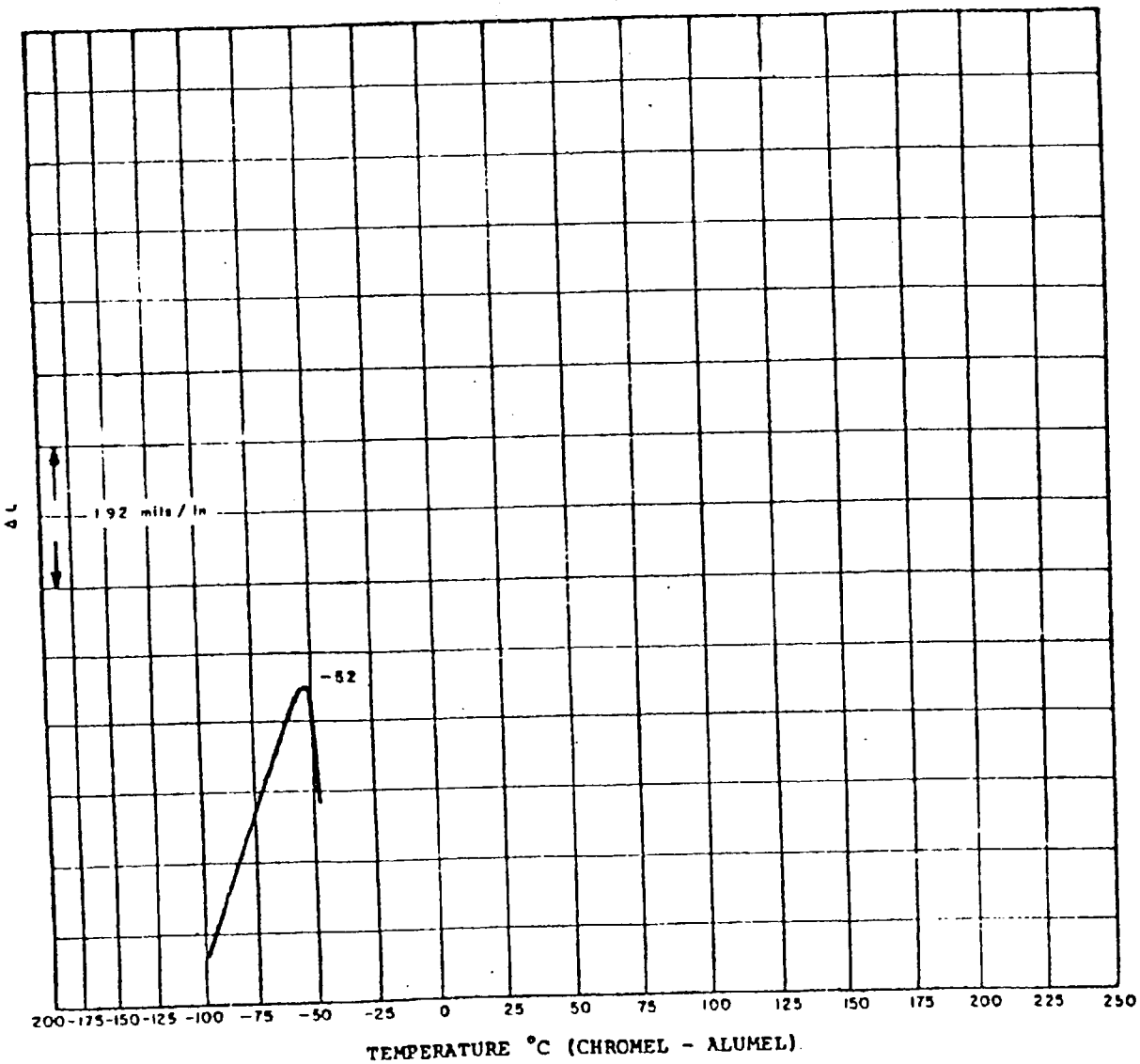


FIGURE 3. Silicone sponge rubber transition temperature by thermomechanical analysis (medium grade).

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-63172 SPACER, GRAIN M753 8 INCH PROJECTILE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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